CFFM-Net: Reliability-Gated Mamba Fusion of Visible and Thermal Images for Detection, Tracking and Multi-Camera Analytics

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Abstract

Visible-thermal detectors usually fuse their two cameras as if both were always healthy, yet in service one camera is often dark, glared, misaligned or gone. CFFM-Net is a dual-stream YOLO26 detector whose fusion block estimates a per-pixel reliability for each camera and uses it to scale the discretisation step of a four-direction selective state-space scan. At nano scale and over 3 seeds, it beats a matched concatenation baseline on M3FD by $+0.6 \pm 0.4$ mAP (paired $p = 0.11$, not significant) and is level on LLVIP ($+0.8 \pm 1.1$, $p = 0.37$) and KAIST. Fusion beats the best single camera on M3FD by +4.0 mAP, of which the baseline captures +3.4. On an eleven-fault benchmark the model leads on thermal-side faults on M3FD (+3.6 pp) and trails on visible-side ones (−0.9 pp), but a baseline trained with fault augmentation is more robust than the model. It costs 1.36 times the baseline latency. A Temporal State Memory lowers the KAIST miss rate under thermal noise from 51.6 to 34.1 %, with a confound we state, but not when the thermal camera is lost or in tracking. A live runtime keeps delay and memory bounded over ten minutes, at 12.4 frames per second on a nominal 30. An RGB-only multi-camera layer on WILDTRACK reaches 77.4 % MODA.

Keywords: Multispectral object detection, visible-thermal fusion, state space models, Mamba, sensor faults, multi-object tracking

1. Introduction

At night a visible camera sees little and a thermal camera sees warm bodies without texture, so pairing the two is the standard answer for pedestrian and vehicle detection after dark. The pairing carries a cost that benchmark tables hide. Public paired datasets are curated so that both streams are present, registered and healthy in every frame. A deployed pair is none of those things for long. The visible camera is blinded by headlights or loses its exposure in a tunnel, the thermal core stalls during a shutter calibration, a lens knock shifts one image against the other, and a cable pull removes a stream entirely. A fusion network trained only on clean pairs has never been asked what to do when one input is wrong, and a detector that falls below its single-camera baseline in that moment is a worse sensor than either camera alone.

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Recent fusion detectors build on selective state-space models [1], which mix tokens along a sequence in linear time. Fusion-Mamba and CFMW scan the two modalities jointly and weight them afterwards [2, 3]; transformer fusers such as CFT and ICAFusion attend across modalities [4, 5]. As far as their descriptions show, in each case the combination rule acts on features that a faulty camera has already shaped. In a selective scan the shaping happens early, because every token writes into a recurrent state through its own step size, and a corrupted token writes just as strongly as a clean one.

CFFM-Net (Cross-Feature Fusion Mamba Network) moves the decision to where the write happens. A small head predicts a reliability in $(0, 1)$ for each camera at each location, and the scan multiplies the step size of a token by the reliability of the camera it came from. A token whose reliability is zero would have a zero step and leave the state untouched; in practice the fusion weight of a round-one model stays close to one half (Section 4.8), so the learned reliability acts as a mild prior. The fusion block sits on top of a stock YOLO26 detector [6], so the trainer, loss and evaluation are unchanged, and it is small enough to keep the whole detector at nano scale.

We present the detector, its baselines, ablations and a sensor-fault benchmark, a temporal extension evaluated on video, a live runtime, and the system built around them. The evidence is mixed in places, and we report it that way. Over three seeds the model beats a matched concatenation baseline on M3FD by $+0.6 \pm 0.4$ mAP, which is not significant at $n = 3$ (paired $p = 0.11$), and is level with it on LLVIP and on KAIST still images. Fusing the two cameras beats the best single camera on M3FD by +4.0 mAP, but the concatenation baseline captures +3.4 of it. On the fault benchmark a baseline trained with fault augmentation is more robust than our model, which limits what the

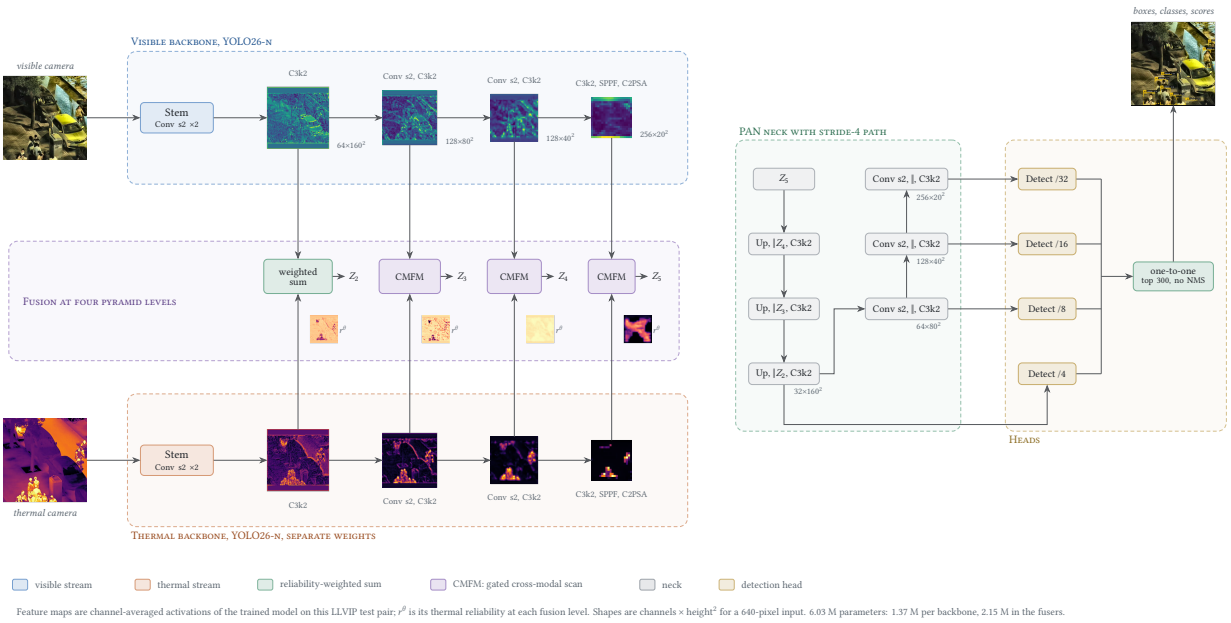


Figure 1: CFFM-Net v1 on a LLVIP pair. Two YOLO26-n backbones with separate weights are fused at each pyramid level; the neck and head are stock YOLO26. The thermal reliability r^θ is shown at each level. Versions v2 and v3 drop the stride-4 level and its detection head.

reliability gate can claim. In a preliminary single-seed run, a later variant with an auxiliary reliability loss is worse than the plain model, so the loss does not earn a place in the recipe. The contributions are as follows.

1. A cross-modal fusion block, CMFM, that gates the discretisation step of a four-direction selective scan by per-pixel, per-camera reliability, with a learned offset alignment and a convolutional local branch, built as a drop-in extension of YOLO26 (Section 3).
2. A controlled evaluation on LLVIP and M3FD with a matched baseline, three seeds, matched single-camera baselines on M3FD, size-banded average precision, and an eleven-fault sensor benchmark that includes a fault-trained baseline and the negative ablation of the reliability loss, set beside published results with their protocol differences stated (Section 4).
3. A Temporal State Memory that carries a per-pixel scan state across video frames, warped by frame-to-frame motion and slowed by the sensor reliability, and a live runtime with a stream health monitor that feeds sensor flags to the model (Sections 3.7 and 3.9), evaluated on KAIST (Section 5) and in a soak and link-fault test on replayed streams (Section 6), where the memory helps under thermal noise and visible darkness and not under thermal loss or in tracking.
4. A multi-camera analytics layer on a ground plane, shown as an RGB-only system demonstration on WILDTRACK with a plain RGB detector and no fusion (Section 7).

The remainder of the paper reviews related work, describes the method, reports the detection, video and live-runtime experiments, describes the system, and discusses limits. Section 8 lists what the evidence does and does not support.

2. Related Work

2.1. Real-time detectors and small objects

Real-time detection rests on feature pyramids [7] and path aggregation [8], and the YOLO family, most recently YOLO26 [6] and its predecessor YOLO11 [9], packages both with a one-stage head. Small objects remain the acknowledged failure case. HPS-DETR reports that objects below ten pixels and distant objects are still missed [10], and CFMW reports that small objects are still its weak point [3]. We therefore report average precision by size band, rather than a single mAP, throughout.

2.2. State-space models in vision

Mamba makes the state-space step input dependent [1]. Vision variants differ mainly in how they order two-dimensional tokens: bidirectional scans [11], cross scans over four directions [12], and hybrids that place Mamba and attention at different depths [13]. Mamba YOLO brings the idea into a YOLO detector [14], and UAVDet pairs convolution with Mamba for small objects in aerial imagery [15]. UAVDet states that the sequential state update uses GPU parallelism poorly, and CFMW reports a large rise in computation from its scan. Our scan uses the four-direction order of VMamba and keeps the state small (state size 8), and our cost results in Section 4.6 quantify the price.

2.3. Visible-thermal object detection

Multispectral pedestrian detection grew up on KAIST [16], where early work compared where in a network the two streams should meet [17] and added a segmentation branch to the detector [18]. Later work in this line addressed misalignment between the two cameras, with a weakly aligned cross-modal learner [19], and built interaction and guidance modules on two-stream convolutional detectors, namely cross-modality interactive attention [20], guided attentive fusion [21] and channel switching with spatial attention [22]. More recent detectors on KAIST, LLVIP and M3FD fall into three groups. Transformer fusers attend across the two modalities inside a YOLO-style or two-stage detector: CFT [4] and ICAFusion [5], and IRDFusion with iterative relation-map differences [23]. Detection transformers adapt the DETR family to two inputs, as in MS-DETR [24], DAMSDet [25] and GM-DETR [26]. Other designs change the pipeline around the detector: a single-stage multispectral framework built on YOLOv11 [27], a target-aware fusion for pedestrians [28], an end-to-end design that trains image fusion and detection together [29], and a coarse-to-fine removal and selection scheme [30]. Remote-sensing detectors such as SuperYOLO and its relatives pair multispectral inputs with YOLO-style heads [31, 32]. We compare with the published numbers of several of these in Section 4.7 and do not re-run them.

State-space fusion has appeared in this setting recently. Fusion-Mamba builds cross-modality fusion modules from state-space blocks [2], CFMW targets adverse weather [3], WaveMamba applies a wavelet-driven Mamba fusion [33], and DMM uses disparity guidance for oriented objects in remote sensing [34]. As far as their titles and abstracts show, these designs use the scan to mix the two modalities; we did not find one that gates the step of the scan by a per-camera reliability. MambaST fuses a few consecutive frames of a visible-thermal pair for pedestrian detection [35]; our Temporal State Memory differs in carrying a recurrent state indefinitely and in tying its decay to sensor reliability and real elapsed time.

2.4. Reliability, illumination and modality imbalance

The idea that one camera can be less trustworthy than the other is not new. MBNet argues that multispectral detectors suffer from modality imbalance and uses illumination information to align and weight the two feature streams [36], and MS-DETR adds a modality-balanced optimisation to its loosely coupled fusion [24]. Explicit attention across the two modalities has been proposed for several RGB-thermal tasks [37], and probabilistic ensembling fuses the outputs of per-modality detectors [38]. In the works we surveyed, reliability or illumination weights the modalities on the features or on the output of the fusion. We found no method that controls the discretisation step of the scan itself per location and per modality, which is the design choice we test. In the works we surveyed we also found no benchmark that corrupts one camera of a pair with a range of physical faults, which is why Section 4 builds one.

2.5. Datasets and tracking

LLVIP provides registered visible-infrared pairs for low-light pedestrian detection [39], and M3FD offers multiple scenarios and classes [40]. KAIST is the classic multispectral pedestrian benchmark, with sanitised annotations

available [41], and its numbers depend strongly on whether training and test annotations are original or sanitised [21]. A bag-of-tricks study benchmarks training choices for multispectral detectors [42]. WILDTRACK supplies calibrated, synchronised multi-camera video of pedestrians [43]. For tracking by detection we use ByteTrack, which associates low-confidence boxes as well as confident ones [44].

3. Method

3.1. Detector and notation

CFFM-Net extends YOLO26-n. A pair enters as a four-channel tensor, visible BGR and one thermal grey channel, and the thermal channel is repeated to three for its backbone. Two backbones with separate weights, both initialised from COCO-pretrained YOLO26-n, produce feature maps at each pyramid level that the head reads. A fuser at each of those levels combines the visible map F^v and the thermal map F^θ into one map Z of the same shape, and the stock YOLO26 neck and one-to-one head consume the fused maps. Because the model subclasses the Ultralytics detection model and keeps the loss and validator unchanged, every result below comes from the standard trainer and the standard COCO-style evaluation. Figure 1 shows the layout.

Feature maps have c channels and $h \times w$ cells at the level in question. Reliabilities $r^v, r^\theta \in (0,1)^{h \times w}$ are single-channel maps.

3.2. Reliability head

For each level, a head predicts how far each camera can be trusted at each cell from the evidence in both maps:

$$[r^v, r^\theta] = \sigma\Big(W_r\, \phi([F^v,\ F^\theta,\ |F^v - F^\theta|])\Big), \tag{1}$$

where ϕ is a depthwise 3×3 convolution with batch normalisation and SiLU, W_r is a 1×1 convolution to two channels, and σ is the logistic function. W_r starts at zero weight with a bias of 2, so both cameras are trusted at initialisation. The absolute difference gives the head a direct view of disagreement between the cameras. An optional sensor flag $f \in \{0,1\}^2$ per sample multiplies the output, which is how an external monitor can declare a camera dead: the flag forces that camera's reliability to zero everywhere.

3.3. Offset alignment

Thermal and visible images are registered in LLVIP but not perfectly in the field. Before the scan, a small convolutional head predicts a two-channel displacement $\Delta = \delta_{\max} \tanh(g([F^v, F^\theta]))$ with $\delta_{\max} = 4$ cells, and bilinear sampling resamples the thermal map at the displaced positions, giving $\tilde{F}^\theta$. The output layer of g starts at zero, so the block starts as the identity warp and learns a shift only if the loss rewards it.

3.4. Reliability-gated cross scan

Both maps are projected to $d = \max(16, \mathrm{round}(0.5\,c))$ channels by a 1×1 convolution followed by a depthwise 3×3 convolution, giving p^v and p^θ. At every cell the visible and thermal tokens are placed next to each other, so a sequence of $L = 2hw$ tokens alternates $p^v_1, p^\theta_1, p^v_2, p^\theta_2, \ldots$ over the raster order. The block scans this sequence in $K = 4$ orders: rows forward and backward, columns forward and backward. For one direction, token u_ℓ with modality $m(\ell) \in \{v, \theta\}$ and grid position $\mathrm{pos}(\ell)$, the recurrence is

$$\Delta_\ell = \mathrm{softplus}(W_\Delta u_\ell + b_\Delta) \cdot r^{m(\ell)}_{\mathrm{pos}(\ell)}, \tag{2}$$

$$h_\ell = e^{\Delta_\ell A}\, h_{\ell-1} + \Delta_\ell\, B_\ell\, u_\ell, \tag{3}$$

$$y_\ell = C_\ell\, h_\ell + D\, u_\ell, \tag{4}$$

with A a diagonal matrix of negative entries (state size 8), and B_ℓ, C_ℓ and the pre-activation of Δ_ℓ produced from the token by learned projections, as in Mamba [1]. The only change from a standard selective scan is the last factor in the first line, the reliability at the token's position for the token's own camera. When that reliability is zero, $\Delta_\ell = 0$, the decay factor is one, the input term vanishes, and the state passes the token untouched. A camera with zero reliability therefore cannot alter what the scan carries forward. The outputs of the four directions are mapped back to the grid and summed (Fig. 2).

3.5. CMFM and fusion levels

The Cross-Modal Fusion Mamba block (CMFM) wraps the scan with two further paths. The scanned outputs of both modalities are concatenated and normalised by a channel LayerNorm. A local branch applies a depthwise 3×3 convolution to the concatenation of p^v and p^θ and restores the two-dimensional neighbourhood that a one-dimensional scan weakens. A 1×1 convolution on the concatenation of both paths gives a residual M, and the block outputs

$$Z = \frac{r^v\, F^v + r^\theta\, \tilde{F}^\theta}{\max(r^v + r^\theta,\, 0.01)} + M. \tag{5}$$

The projection producing M starts at zero, so a freshly built block is exactly a reliability-weighted average of the two maps, and the scan has to earn any change. The weight floor keeps the gradient of the quotient bounded when both reliabilities approach zero. The scan runs in single precision. Three interchangeable scan implementations share one interface: a fused CUDA kernel when available, a fused Triton kernel, and a chunked two-pass PyTorch scan that serves as the reference and runs on any device.

The cost of the scan grows linearly in the number of cells, the number of directions, the channel width and the state size, and it is sequential along each of the $2hw$ tokens, which is why the stride-4 level is excluded from it. The implementation is covered by tests that check four properties: the chunked PyTorch scan equals the textbook recurrence in values and gradients; a token with zero step cannot write into the state; a freshly built CMFM is exactly a reliability-weighted average; and the offset alignment is the identity at initialisation.

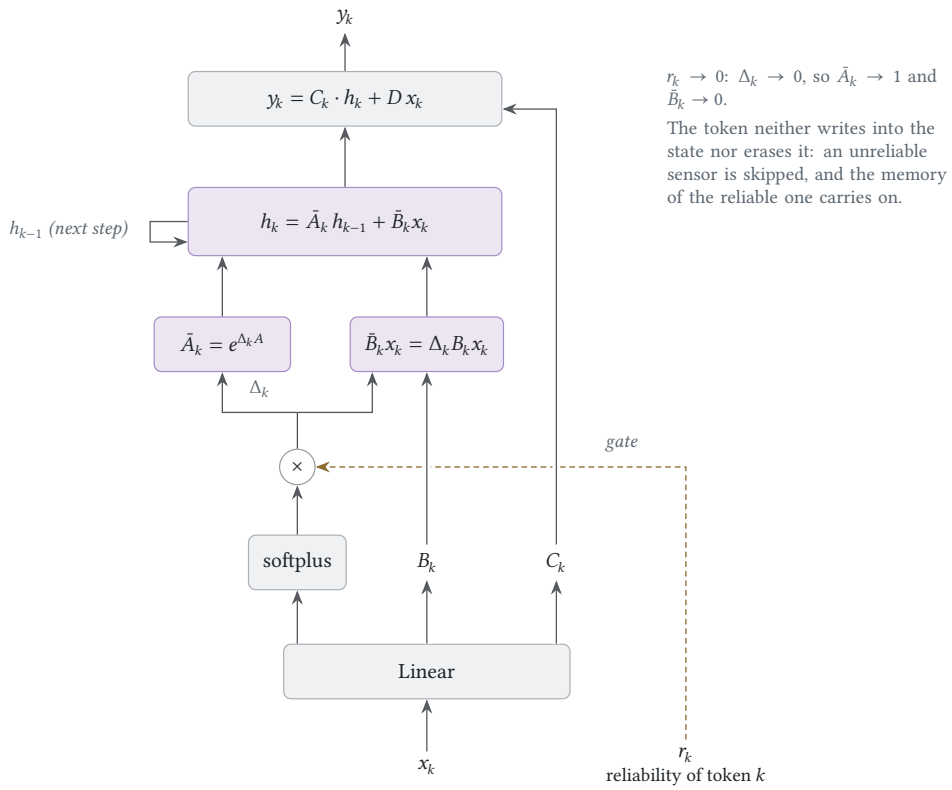


Figure 2: One step of the gated scan. The reliability of the token's camera multiplies the step Δ_k before it enters the decay and the input term, so a reliability near zero leaves the state unchanged.

Figure 3 shows the block. The detector fuses at the strides that its head reads. In the first version (v1) this is four levels: a reliability-weighted average at stride 4, where a scan over $2hw$ tokens is expensive and buys little, and CMFM at strides 8, 16 and 32. Versions v2 and v3 read the head at strides 8, 16 and 32 only. The baseline replaces every fuser by concatenation followed by a 1×1 convolution that squeezes the channels back to c, the usual two-stream design.

3.6. Training recipes

Three recipes are compared; they differ in head layout and in what the training batches look like.

- **v1.** Four fusion levels, clean pairs only. It is the round-one model.
- **v2.** Three levels and *modality dropout*: each training pair has probability 0.15 of having its visible camera blacked out, and the same probability for the thermal camera. For half of the blacked-out samples the model also receives the sensor flag. The control baseline is trained with the same dropout, so the comparison isolates the fuser.

- **v3.** v2 plus *fault training* and a *reliability loss*. With probability 0.3 a sample has one random camera corrupted by one of the faults of the benchmark below at a random severity, and the head receives a trust target for that camera of 0.7, 0.4 or 0.15 at severity one, two or three, and zero for a blacked-out camera (healthy cameras have target one). The reliability loss is the binary cross-entropy between the mean reliability of each gated fuser and its trust target, averaged over fusers and added with weight 0.5. The ablation *v3 without reliability branch* keeps the fault training and removes the loss; the *fault-trained concat* baseline keeps the fault training and the concat fuser.

3.7. Temporal State Memory

A still-image detector forgets a pedestrian the moment its thermal camera drops a frame. The Temporal State Memory (TSM) gives each fusion level a per-pixel recurrent state that survives across frames. It is attached after the fuser at each level with a residual path that starts at zero, so a model with a fresh memory produces exactly the detections of the model without it (Fig. 4).

At frame t the fused map Z_t at one level is projected to $c_s = \mathrm{round}(0.5\,c)$ channels, $u_t = W_u Z_t$, and every channel at every cell keeps a state of size 8. The update is a selective scan in time rather than space:

$$\delta_t = \mathrm{softplus}(W_\delta u_t) \cdot \frac{\tau_t}{\tau_0} \cdot \rho_t, \tag{6}$$

$$S_t = e^{\delta_t A} \odot \mathcal{W}_{H_t}(S_{t-1}) + \delta_t\, B_t\, u_t, \tag{7}$$

$$y_t = C_t\, S_t + D\, u_t, \qquad Z'_t = Z_t + W_o\, \mathrm{DW}(y_t), \tag{8}$$

where B_t and C_t come from u_t by a 1×1 convolution, DW is a depthwise 3×3 convolution spreading the readout to neighbours, and W_o is zero-initialised. Three factors act on the step. The ratio τ_t/τ_0 of the real time gap since the previous frame to a nominal gap $\tau_0 = 0.1$ s makes the decay depend on elapsed time, so a stalled camera that returns after a long gap forgets in proportion. The reliability $\rho_t = \max(r_t^v, r_t^\theta)$ from CMFM is high when at least one camera is trusted, and a frame in which neither camera is reliable writes little into the state. The warp $\mathcal{W}_{H_t}$ resamples the previous state under the homography H_t from frame $t-1$ to t, with zeros for newly exposed cells, so the memory follows a moving camera rather than smearing. A scene cut, or a time gap beyond 2 s in the live runtime, resets the state.

The memory is trained with the detector frozen. Clips of 6 frames at stride 2 are drawn from KAIST training sequences [16], with labels for every frame. With probability one half a clip receives a fault, one of drop, darkness, noise, glare or shift, starting at a random frame and lasting to the end of the clip, which is the situation the memory exists for. The loss is the detection loss averaged over the frames of the clip, backpropagated through time, optimised with AdamW under a one-cycle schedule, with gradient clipping and with any non-finite step skipped.

3.8. Causal inference and tracking on video

Video inference is causal: frames are processed in order, the state is reset at the start of each sequence, and no future frame is used. For tracking, detections above a low confidence threshold feed ByteTrack [44], which also associates low-score boxes with existing tracks. Because the memory affects detections only, switching it off at test time gives the same model without memory, which is the ablation for video. A scheduler can inject a fault into a sequence, healthy for a number of frames and failed for the next number, repeating, to test recovery in the way a flaky link would behave. Detection is scored on KAIST by the log-average miss rate in the standard reasonable setting, split into day, night, failed and healthy frames, and tracking is scored by MOTA, IDF1 and identity switches against the KAIST person tracks.

3.9. Live runtime

The live runtime turns the model into a process that reads two cameras and keeps running when they misbehave.

Capture. Each camera runs in its own thread, stamps every frame with a monotonic time at the moment of capture, and writes it into a one-slot queue. A put replaces the frame waiting there and counts it as dropped, so a slow consumer never builds a backlog and latency cannot grow with time. A source that fails is reopened with backoff. Sources are a webcam index, an RTSP stream over TCP, or a replay of a recorded sequence at real speed with injected link faults: stalls, frozen frames, unplugs, random drops, delay and jitter.

Health monitor. A monitor watches each camera from arrival times and pixels. It declares a camera stalled when no frame has arrived for 4 nominal frame intervals, frozen when a downsampled frame has stayed nearly unchanged for 6 consecutive frames, black when the mean intensity is near zero, and saturated when a large share of pixels sits at the top of the range. Stalled, frozen and black cameras are *down*, a saturated camera is *degraded*, and others are *ok*. The nominal rate is 30 frames per second.

Inference loop. Each step takes the newest visible frame, waiting at most one frame interval, and the newest thermal frame without waiting, and reuses the last frame of a camera that has produced none. A camera that is down contributes a zero image and a zero sensor flag, which forces its reliability to zero everywhere, so the detector sees the failure as a stated fact rather than inferring it from a black picture. The gap between the timestamps of consecutive newest frames sets τ_t for the memory, and a gap longer than 2 s resets it. The model runs under half precision, followed by ByteTrack. Each step records inference time, end-to-end time from capture of the newest frame to the end of tracking, the time gap, the offset between the two camera timestamps, the health of each camera and the allocated GPU memory.

Summaries. A run is summarised by throughput, median and 95th-percentile inference and end-to-end time, drift (the median end-to-end time over the last fifth of the run minus that over the first fifth), and growth in GPU memory. Fault handling is summarised by the delay from a fault's onset to the first *down* status, the delay from its end to the first *ok*, and the rate of false alarms.

4. Experiments

4.1. Setup

Data. LLVIP [39] is a registered visible-infrared set of pedestrians taken at night; we train on every 3rd frame of the training split and evaluate on the full test split. M3FD [40] has multiple scenarios and classes and no official detection split, so we hold out a fixed random 20 % of the pairs with a seeded split that is shared by every run. Both datasets are converted to a four-channel paired layout, with images resized to 640 pixels.

Models and training. The baseline is the two-stream concatenation detector with the same backbones, neck and head, and matched training. Every fusion run uses COCO-pretrained YOLO26-n weights in both backbones, batch size 16, and the stock Ultralytics trainer. Round one (v1) trains for 30 epochs on clean pairs and is run only for seed zero. All later rounds train for 80 epochs with mosaic augmentation switched off for the last 10. The headline comparison, v2 against the concat baseline with modality dropout, is run with 3 seeds on both datasets. The v3 family, its ablation and the fault-trained concat baseline are run with seed zero only, so v3 has no spread. The single-camera baselines are YOLO26-n detectors on one image stream, one seed. On M3FD they use the same pretrained weights, 80 epochs and mosaic schedule as the fusion runs and a batch size of 32 instead of 16; on LLVIP they were trained for 30 epochs and are not matched. Tables report percent as mean ± sample standard deviation over seeds ($n = 3$), and margins are means of per-seed differences, in percentage points (pp). A printed margin can differ by a tenth from the difference of the printed means because the per-seed differences are not rounded first.

The five hypotheses tested in this work and their decision thresholds were fixed in advance, in the project plan of the repository, before the experiments ran.

Metrics. We report COCO-style mAP at IoU 0.5 to 0.95 (mAP) and at 0.5 (mAP_{50}), and average precision for the COCO small, medium and large size bands. For M3FD we also report three finer bands from the AI-TOD convention, very tiny, tiny and small, which separate the smallest objects. Latency is batch one with 512×640 input under half precision, after 50 warm-up iterations and over 300 timed iterations, from CUDA events.

Sensor-fault benchmark. Eleven physically motivated faults are applied to one camera of the test pairs, each at three severities. The visible camera suffers darkness, glare, fog, noise, blur and a full drop; the thermal camera suffers a full drop, a spatial shift against the visible image, column banding with pixel noise, loss of contrast toward the scene mean (thermal crossover), and dead pixels. Severity three is the harshest and is the one reported. A drop has no severity, so its three levels coincide. In two further rows the dropped camera is declared to the model by the sensor flag. The benchmark scores v2 and the concat baseline with the checkpoints of all three seeds on M3FD and with the seed-zero checkpoints on LLVIP, whose seed-one and seed-two faults were not run. The fault-trained concat baseline is scored at seed zero on both datasets. Severity three is the one tabulated, and a seed contributes one value per fault.

4.2. Main results

Table 1 gives the headline comparison. On M3FD, CFFM-Net v2 reaches 53.9 ± 0.1 mAP against 53.3 ± 0.3 for the concat baseline, a margin of $+0.6 \pm 0.4$ pp, and v2 is ahead on every one of the three seeds (Table A.14). The margin is small and not significant at $n = 3$: the paired t-test gives $p = 0.11$ and the 95 % confidence interval of the margin is -0.3 to $+1.5$ pp ($t_{0.975,2} = 4.303$). Three wins out of three seeds have a one-sided sign-test probability of 0.125 under no difference (0.25 two-sided), so the consistent sign is weak evidence. At IoU 0.5 the M3FD margin is $+0.9 \pm 0.7$ pp.

On LLVIP the two models are level: 66.2 ± 0.8 for v2 against 65.5 ± 0.8, a margin of $+0.8 \pm 1.1$ pp whose spread exceeds its mean (paired $p = 0.37$, 95 % confidence interval -2.1 to $+3.6$ pp). v2 is ahead on two of three seeds. We therefore read LLVIP as a tie and do not claim an LLVIP gain. LLVIP has one class and a thermal camera that already separates people from background, which leaves little room for a better fuser.

The single-camera rows give context. On M3FD the matched baselines reach 49.9 mAP on the visible images and 47.6 on the thermal images, so the visible camera is the stronger single camera there. Fusion beats it by a clear margin. CFFM-Net v2 is $+4.0 \pm 0.1$ pp above the visible-only detector (+3.8 pp on its lowest seed) and the concat baseline is $+3.4 \pm 0.3$ pp above it. At IoU 0.5 the v2 gain is +4.7 pp, and on small objects +5.0 pp. Most of the gain from fusing the two cameras on M3FD is thus available to the plain concat baseline, and the part that v2 adds is the margin of the previous paragraphs. Two caveats apply. The single-camera values are one seed, so the spread shown is that of the fusion side only and the true uncertainty of the gain is larger, and the single-camera batch size is twice the fusion batch size. On LLVIP the single-camera rows are not matched: 52.0 mAP for the visible images and 64.1 for the thermal ones, at 30 epochs and one seed against 80 epochs for the fusion rows, so they support no conclusion about the gain from fusion there. At the round-one budget the concat baseline reaches 64.9 and CFFM-Net v1 62.9, and the round-one hypothesis that fusion beats the best single sensor at night came out inconclusive (-0.0120 AP vs thermal-only).

4.3. Small objects

Table 2 separates the size bands. On M3FD, v2 gains on small objects, $+0.4 \pm 0.3$ pp, medium objects are level (0.0 ± 0.9 pp), and large objects gain $+0.5 \pm 1.2$ pp with a spread larger than the mean. The small-object margin is the only band whose spread is smaller than its mean, and at $n = 3$ it is not significant. The finer bands in Table B.16 point the same way for the smallest objects, though the very tiny band is noisy: v2 reaches 8.0 ± 0.8 against 7.6 ± 0.2, and in the tiny band 24.9 ± 0.3 against 23.9 ± 2.1, whose baseline spread is wide.

On LLVIP the picture reverses at the small end. The margin of v2 on small objects is -0.4 pp (spread 0.5), while it gains +0.8 and +0.8 pp on medium and large ones. We do not claim a small-object advantage on LLVIP. The round-one model also failed the small-object hypothesis fixed in advance on M3FD: -0.0248 AP in the 8-32 px bands, verdict against. The v2 gain therefore arrives only after the changes of Section 3.6, and the evidence for it is one dataset.

Table 1: Detection accuracy in percent. Fusion models: mean and sample std over three seeds, 80 epochs. Single-camera rows: one seed; 30 epochs on LLVIP, which is not matched to the fusion rows, and 80 epochs on M3FD, which is. The margin over the best single camera subtracts the one-seed single-camera value from each fusion seed.

Model	mAP	mAP_{50}
LLVIP		
YOLO26-n visible	52.0	90.0
YOLO26-n thermal	64.1	96.2
Concat, modality dropout	65.5 ± 0.8	96.8 ± 0.4
CFFM-Net v2	66.2 ± 0.8	96.9 ± 0.3
Margin of v2 over concat (pp)	$+0.8 \pm 1.1$	$+0.2 \pm 0.5$
M3FD		
YOLO26-n visible	49.9	78.0
YOLO26-n thermal	47.6	75.0
Concat, modality dropout	53.3 ± 0.3	81.9 ± 0.3
CFFM-Net v2	53.9 ± 0.1	82.7 ± 0.6
Margin of v2 over concat (pp)	$+0.6 \pm 0.4$	$+0.9 \pm 0.7$
Margin of v2 over best single camera (pp)	$+4.0 \pm 0.1$	$+4.7 \pm 0.6$

4.4. Sensor faults

Table 3 reports the benchmark. On M3FD every number for v2 and the modality-dropout concat baseline is a mean over three seeds; on LLVIP it is one seed, and seed-to-seed spread of clean mAP there is about one point (Table A.14), so LLVIP differences of one to three points are within seed noise. Faults on one camera cost every detector a large part of its mAP. Averaged over the eleven faults, v2 keeps 76.3 ± 0.4 % of its clean mAP on M3FD against 75.0 ± 0.2 % for the concat baseline, and 70.3 % against 75.5 % on LLVIP, where the baseline retains more. A thermal shift is the worst fault for every model on both datasets, and it leaves v2 2.8 mAP on LLVIP and 26.9 on M3FD.

The comparison between v2 and the concat baseline depends on the camera that fails (Fig. 5). On M3FD, v2 is ahead on every thermal-side fault: dead pixels by +5.5 pp, noise by +7.6, shift by +2.5, crossover by +1.4 and a thermal drop by +0.9. It is behind on every visible-side fault, from −0.2 (darkness) to −2.3 pp (noise). Averaged over the faults of each camera the margin is $+3.6 \pm 0.2$ pp for thermal faults and -0.9 ± 0.9 pp for visible faults, and over all eleven it is $+1.2 \pm 0.5$ pp (paired $p = 0.06$ over the three seeds, 95 % interval −0.2 to +2.5 pp), which is not significant at $n = 3$. The sign of the per-fault margin agrees on all three seeds for 6 of the 11 faults. The thermal-side advantage is consistent in direction, but the seed spread of single faults is large (for example ±3.8 pp on thermal noise), so we read it as a pattern the data support without a significance test passed. It is plausible for the design: when the thermal stream is degraded, the gate can lower its weight and the scan writes less of it into the state, while visible-side faults are corrupted before the visible features exist.

On LLVIP, with one seed, the benchmark is less kind. v2 is ahead on 7 of the eleven faults, including darkness (+1.4 pp), glare (+1.4), fog (+0.8) and a visible drop (+1.3), but it loses heavily to additive noise on either camera,

Table 2: Average precision by object size in percent, mean and sample std over three seeds, 80 epochs.

Model	Small	Medium	Large
LLVIP			
Concat, modality dropout	9.6 ± 1.5	65.0 ± 0.6	67.9 ± 1.6
CFFM-Net v2	9.1 ± 1.0	65.8 ± 0.6	68.7 ± 1.9
Margin of v2 (pp)	-0.4 ± 0.5	$+0.8 \pm 1.1$	$+0.8 \pm 1.4$
M3FD			
Concat, modality dropout	34.8 ± 0.2	66.8 ± 0.8	87.2 ± 1.1
CFFM-Net v2	35.2 ± 0.3	66.7 ± 0.2	87.7 ± 0.3
Margin of v2 (pp)	$+0.4 \pm 0.3$	0.0 ± 0.9	$+0.5 \pm 1.2$

by −8.1 pp on the visible side and −18.8 on the thermal side, and to dead thermal pixels (−9.3). The mean margin on LLVIP is −2.9 pp. The gate is not a general robustness mechanism: it helps on specific faults, the thermal-side ones on M3FD and the exposure ones on LLVIP, and it hurts when noise is injected on LLVIP.

The strongest result of the benchmark concerns neither fusion design. A concat baseline trained with fault augmentation (Concat FT in Table 3, seed zero) retains 86.8 % of its clean mAP on M3FD and 87.3 % on LLVIP, against 76.3 % and 70.3 % for v2, and its mean mAP over the faults is higher than that of v2 by 5.1 pp on M3FD and 11.9 pp on LLVIP, while its clean mAP is within one point of v2 (v2 minus Concat FT: +0.8 pp on M3FD and −0.8 pp on LLVIP). Training on faulty inputs buys more robustness than the reliability-gated scan does, at no cost in clean accuracy. We did not run the fault benchmark on v3, which combines the scan with fault training, so we cannot say whether the scan adds anything once the training is matched; this is the comparison that would settle the value of the gate for robustness.

Telling the model which camera dropped through the sensor flag changes nothing measurable. The flagged and unflagged rows of Table 3 differ by at most two tenths of a point for every model, which suggests that detectors trained with modality dropout already recognise a black input without being told.

4.5. Ablations and the negative result

Table 4 lists every variant and Table 5 the margins over the control in each round. Three findings stand out.

First, round one was negative. CFFM-Net v1 had a margin of −2.1 pp on LLVIP and −2.6 pp on M3FD against the concat baseline. Part of the gap came from the stride-4 level: adding that head lowers the concat baseline itself from 64.9 to 63.2 on LLVIP, and against the concat baseline with the same head, the margin of v1 is −0.3 pp. We took that as the reason to drop the stride-4 level in v2, a decision made on LLVIP at seed zero, since the P2 ablations were not run on M3FD. Removing the gate costs v1 +0.3 pp and replacing the scan by a gated convolution of matched cost +0.1 pp, both within the noise of one seed. We therefore do not claim that the scan or the gate, on their own, explain the v2 gain; the v2 comparison is between complete systems.

Table 3: Fault benchmark at severity three, mAP in percent, 80 epochs. LLVIP is seed 0 only. On M3FD the v2 and concat columns (modality dropout) are mean and sample std over three seeds, and the fault-trained concat column (Concat FT) is seed 0. Flagged rows tell the model which camera dropped.

	LLVIP, seed 0			M3FD, mean ± std, three seeds		
Fault	v2	Concat	Concat FT	v2	Concat	Concat FT
Clean	65.6	64.8	66.4	53.9 ± 0.1	53.3 ± 0.3	52.9
Visible dark	63.0	61.6	63.4	41.0 ± 0.9	41.1 ± 0.8	42.7
Visible glare	63.6	62.2	65.0	43.4 ± 0.5	43.8 ± 0.6	46.2
Visible fog	63.1	62.3	64.5	48.0 ± 0.7	49.0 ± 0.4	51.6
Visible noise	53.4	61.5	63.6	39.5 ± 0.9	41.7 ± 1.0	47.1
Visible blur	60.6	62.1	64.6	38.6 ± 1.3	39.8 ± 0.6	43.8
Visible drop	63.3	62.0	63.4	41.5 ± 0.7	41.8 ± 0.7	41.4
Thermal drop	48.3	48.2	47.0	46.2 ± 0.5	45.3 ± 0.2	45.6
Thermal shift	2.8	2.8	24.0	26.9 ± 0.4	24.4 ± 0.4	39.3
Thermal noise	22.3	41.1	59.6	39.2 ± 0.7	31.6 ± 3.4	48.2
Thermal crossover	41.7	41.0	61.4	47.6 ± 0.9	46.2 ± 0.8	49.7
Thermal dead	24.5	33.8	61.0	40.3 ± 1.1	34.9 ± 2.6	49.7
Visible drop, flagged	63.1	62.0	63.4	41.5 ± 0.7	41.8 ± 0.7	41.4
Thermal drop, flagged	48.3	48.2	47.0	46.1 ± 0.4	45.3 ± 0.2	45.6
Mean over faults	46.1	49.0	58.0	41.1 ± 0.3	40.0 ± 0.3	45.9
Retained (%)	70.3	75.5	87.3	76.3 ± 0.4	75.0 ± 0.2	86.8

Second, v2 recovered the deficit. With three levels and modality dropout, shared by the baseline, v2 is level on LLVIP and ahead on M3FD.

Third, v3 failed in a preliminary, single-seed result. Adding fault training and the reliability loss gives 64.0 mAP on LLVIP and 51.9 on M3FD, below v2 on both (-1.5 and -1.8 pp) and below the fault-trained concat baseline (-2.4 and -0.9 pp). The informative control is v3 without the reliability branch, which keeps the fault training and removes the loss. It reaches 67.2 on LLVIP and 53.3 on M3FD and so v3 minus this control is -3.2 and -1.4 pp: the branch costs accuracy. We therefore report the reliability loss as a negative ablation and do not include it in the recommended recipe. These are single-seed runs, so the size of the effect is uncertain, but the sign agrees on both datasets. A plausible cause is that the loss pushes the mean reliability toward a target that the detection loss does not need, and the two objectives compete. We did not test that explanation.

4.6. Cost

Table 6 gives the cost of the evaluated models on a RTX 4050 Laptop GPU, in half precision at batch one, timed on an otherwise idle GPU. CFFM-Net v2 has 6.02 M parameters and 16.5 GFLOPs, a median latency of 45.0 ms (about 22 frames per second) and a 95th percentile of 54.7 ms. Against the concat baseline this is 1.48 times the parameters, 1.70 times the FLOPs, 1.36 times the median latency and 1.29 times the 95th-percentile latency, so v2 passes the limit

Table 4: Ablation of mAP in percent. A dash marks a run that was not made.

Variant	LLVIP	M3FD
Round one, 30 epochs, seed 0		
Two-stream concat	64.9	50.5
CFFM-Net v1	62.9	47.9
v1, no gate	62.6	–
v1, gated conv	62.8	–
Concat + P2 head	63.2	–
v1, no P2 head	63.5	–
Rounds three and four, 80 epochs, seed 0		
Concat, modality dropout	64.8	53.5
CFFM-Net v2	65.6	53.7
Concat, fault training	66.4	52.9
CFFM-Net v3	64.0	51.9
v3, no reliability branch	67.2	53.3

Table 5: Round history: mAP margin of CFFM-Net over its control in percentage points.

Comparison	LLVIP	M3FD
Round one, v1 against concat (30 epochs)	−2.1	−2.6
Head control, v1 without P2 against concat with P2	+0.3	–
Round three, v2 against concat (seed 0)	+0.7	+0.3
Round three, v2 against concat (three-seed mean)	+0.8	+0.6
Round four, v3 against fault-trained concat	−2.4	−0.9
Round four, v3 against v3 without reliability branch	−3.2	−1.4
Round four, v3 against v2	−1.5	−1.8

of at most twice the baseline latency that was fixed in advance. Against the thermal-only detector the median latency is 1.74 times higher. v3 and its no-branch control have the size of v2, and v3 is 1.16 times the median latency of v2 in this run, so the timing differences between these three models are small and partly noise. The round-one model v1 was timed in an earlier session on hardware that is not recorded, at 1.79 times the baseline; that run is not comparable with Table 6. The gated-convolution control of matched FLOPs was faster than the scan in that run (34.7 against 41.9 ms), which is the price of a sequential scan on this kind of hardware.

4.7. Context with published detectors

The experiments above compare CFFM-Net with a matched baseline and not with published fusion detectors, because published numbers differ in backbone, input size, training schedule, annotations and split. Tables 7 and 8 nevertheless place our results next to those figures, copied from the cited papers and not re-run, so that a reader can see the distance. Our models are nano scale: CFFM-Net v2 has 6.02 M parameters and 16.5 GFLOPs at 640×512, and

Table 6: Parameters (millions), GFLOPs and latency (ms) on a RTX 4050 Laptop GPU, half precision, batch one, 640×512 input, median and 95th percentile per image over 300 timed iterations after 50 warm-up iterations, on an otherwise idle GPU.

Model	Params	GFLOPs	Median	p95
YOLO26-n thermal only	2.50	5.9	25.8	47.5
Concat, modality dropout	4.07	9.7	33.1	42.5
Concat, fault training	4.07	9.7	31.9	41.3
CFFM-Net v2	6.02	16.5	45.0	54.7
v3, no reliability branch	6.02	16.5	50.6	65.6
CFFM-Net v3	6.02	16.5	52.0	66.3
v2 over concat, ratio	1.48	1.70	1.36	1.29

runs at a median of 45.0 ms on a laptop GPU. Most of the published detectors in the tables use YOLOv5-l, YOLOv8-l or ResNet-50 backbones, and several use detection transformers; we do not tabulate their sizes because we did not verify them against the original papers, and we make no claim about relative efficiency.

On LLVIP, with the official split, v2 reaches 96.9 % mAP_{50} and 66.2 % mAP, in the range of published fusion results (mAP from 59.2 to 71.1) and below the best of them. This is with training on one third of the training frames and without a large backbone. The comparison is weaker than it looks. The LLVIP infrared-only baseline with a large YOLOv5 already reaches 67.0 mAP in the dataset paper [39], so thermal imagery alone nearly saturates the benchmark, and a gap of a few points of mAP between two fusion detectors at different input sizes (one row trains at a much larger input than the others) says little about the fuser. On M3FD no row shares our split. Published values of mAP_{50} run from 64.8 to 92.1 across splits of the same dataset, and the same method (CFT) is reported 21.6 points of mAP_{50} apart in two cited papers that use different splits, so we draw no ranking from the M3FD rows.

On KAIST our miss rates are far above the published ones (Table 8): 19.8 % for the concat baseline and 20.3 % for CFFM-Net v2 against 4.47 to 10.62 for the published detectors, which use ResNet or VGG backbones, sanitised training annotations and the standard evaluation code. We attribute part of the gap to the protocol differences in Section 5.1 and part to scale and a short schedule, but we did not run the experiments that would apportion it, and the table should be read as showing that our KAIST detectors are not competitive with the published state of the art. The KAIST experiments serve the video questions of Section 5 and not a leaderboard claim.

4.8. What the gate learns

Figure 7 plots, for LLVIP and M3FD test images, the share of fusion weight given to the thermal camera against the mean brightness of the visible image, for a round-one model. The share moves in the expected direction. In the darkest LLVIP scenes it leans toward thermal and in the brightest M3FD scenes toward visible. The swing is small, and the share stays close to one half everywhere. The gate at round one is therefore a mild prior and not a switch, which fits the modest gains above. Figure 6 shows the trust map on one night scene.

Table 7: Published RGB-T detection results for context, in percent. Numbers are copied from the cited papers, not re-run. Input size or split is stated per row because mAP depends on both. Our models are nano scale, trained on one third of the LLVIP training frames (mean of three seeds), so they are not a like-for-like comparison. On M3FD no row shares a split with ours, and no ranking is claimed.

Method	Detector	Input or split	mAP_{50}	mAP
LLVIP, official split (12,025 / 3,463 pairs)				
YOLOv5-l, infrared only [39]	YOLOv5-l	not stated	96.5	67.0
CFT [4]	YOLOv5 with CFB	640×640	97.5	63.6
CSSA [22]	Faster R-CNN, ResNet-50	640×512	94.3	59.2
MS-DETR [24]	Deformable DETR, ResNet-50	640×512	97.9	66.1
Fusion-Mamba [2]	YOLOv8-l	640×640	97.0	64.3
DAMSDet [25]	DETR, ResNet-50	1024×1024	97.9	69.6
TFDet [28]	YOLOv5-l	not stated	97.9	71.1
WaveMamba [33]	YOLOv8	not stated	98.3	66.0
IRDFusion [23]	Co-DETR	640×640	98.4	70.9
CFFM-Net v2 (this work)	YOLO26-n, 6.02 M	640, every third frame	96.9	66.2
Concat baseline (this work)	YOLO26-n, 4.07 M	640, every third frame	96.8	65.5
M3FD, splits differ between rows and are not comparable				
TarDAL, fused image [40]	YOLOv5	3,500 train, test not stated	80.7	–
DAMSDet [25]	DETR, ResNet-50	3,368 / 831, by scene	80.2	52.9
TFDet [28]	YOLOv5	2,905 / 1,295	64.8	41.0
E2E-MFD [29]	Detectron2, end to end	2,940 / 1,260	91.8	63.8
Fusion-Mamba [2]	YOLOv8-l	split of Liang et al.	88.0	61.9
WaveMamba [33]	YOLOv8	3,360 / 840	92.1	64.4
CFFM-Net v2 (this work)	YOLO26-n, 6.02 M	3,360 / 840, random	82.7	53.9
Concat baseline (this work)	YOLO26-n, 4.07 M	3,360 / 840, random	81.9	53.3

5. Video Experiments on KAIST

This section asks two questions about the video extension of Section 3.7. Does the fusion block help on a pedestrian benchmark with a long history of published results, and what does the Temporal State Memory add on top of a still-image detector, on clean video and while one camera fails? Every number here comes from one training run per model and one memory training run, so no seed spread is available, and we say so wherever a difference is small.

5.1. Protocol

KAIST [16] supplies visible and thermal video at 20 frames per second. We use the standard split: sets 00 to 05 for training and sets 06 to 11 for testing, with sets 00 to 02 and 06 to 08 recorded by day and the rest at night. The still-image detectors train on every 4th frame of the training sets, with persons at least 20 pixels tall as targets. Both detectors, CFFM-Net v2 and the concat baseline with modality dropout, follow the recipe of Section 4 with 25 epochs and seed zero. The test frames are every 20th frame of the test sets, 2262 frames of which 1463 are by day and 799 by night.

Table 8: KAIST log-average miss rate in percent on the reasonable subset (lower is better). Published rows use the improved test annotations and the standard evaluation code and are copied from the cited papers. Our rows use the original annotations for training and testing, every 20th test frame and our own implementation of the protocol, so they are indicative only.

Method	Detector	Train annotations	All	Day	Night
AR-CNN [19]	VGG-16	sanitized	9.34	9.94	8.38
MBNet [36]	ResNet-50	AR-CNN improved	8.13	8.28	7.86
GAFF, original training [21]	VGG-16	original	10.62	10.82	10.14
GAFF, sanitized training [21]	VGG-16	sanitized	6.48	8.35	3.46
ICAFusion [5]	YOLOv5, CSPDarknet53	sanitized	7.17	6.82	7.85
MS-DETR [24]	ResNet-50	sanitized	6.13	7.78	3.18
TFDet [28]	VGG-16	sanitized	4.47	5.22	3.36
Concat baseline (this work)	YOLO26-n	original	19.8	19.9	18.6
CFFM-Net v2 (this work)	YOLO26-n	original	20.3	21.1	18.5

The metric is the log-average miss rate (MR) over nine false-positives-per-image values between 10^{-2} and 10^{0} on the reasonable subset: persons at least 55 pixels tall and at most partly occluded (visible fraction of at least 0.65). Lower is better. We use the original KAIST annotation files for training and testing and our own implementation of the Caltech matching protocol, not the KAIST evaluation toolbox, and we have not applied the sanitised annotations [41] or the improved test annotations used by most recent papers. The miss rates below are therefore internally consistent but are not directly comparable with published figures, and Section 4.7 places them next to those figures with this caveat.

For video the detector runs causally at 10 frames per second (every second frame) and is scored on the same test grid. A failure schedule repeats 4 healthy seconds and 4 failed seconds. We report the miss rate on all frames and on the failed frames alone. Conditions are a lost thermal camera (a black image), a lost thermal camera declared by the sensor flag, a darkened visible camera, and additive noise on the thermal camera, all at severity three. Detector weights are identical in every row; only the memory is switched on or off at test time. Tracking uses ByteTrack at the full 20 frames per second with a detection threshold of 0.1, on four test sequences (two by day, two by night, 175 annotated persons), scored by MOTA and IDF1.

5.2. Still images

Table 9 gives the single-frame results. CFFM-Net v2 does not beat the concat baseline on KAIST. Over all test frames the miss rate is 20.3 % against 19.8 %, a change of +0.5 points that favours the baseline, and the AP_{50} is 83.9 % against 84.6 %. By day the baseline is ahead by +1.2 points of miss rate; at night the two are level (−0.1 points). With one run each and no estimate of run-to-run spread, which on the detection benchmarks of Section 4.2 was about one point of mAP, these differences are best read as a tie with a slight lean to the baseline. We make no claim that the reliability-gated scan improves still-image pedestrian detection on KAIST. Absolute miss rates are high for the reasons

Table 9: KAIST test results on single frames, reasonable subset. Lower is better for the miss rate. One training run per model, 2262 test frames, so no confidence interval is available.

Measure	Concat	CFFM-Net v2	Change
Miss rate, all (%)	19.8	20.3	+0.5
Miss rate, day (%)	19.9	21.1	+1.2
Miss rate, night (%)	18.6	18.5	−0.1
AP_{50}, all (%)	84.6	83.9	−0.7

Table 10: Temporal State Memory against the same detector without memory on KAIST. A camera fails for four seconds in every eight. Failed frames are those inside a failure. Change is memory minus image. One memory training run, one detector.

	All frames			Failed frames		
Condition	Image	Memory	Change	Image	Memory	Change
Miss rate (%), lower is better						
Clean	20.3	19.8	−0.5	–	–	–
Thermal lost	26.2	25.9	−0.3	31.6	32.3	+0.7
Thermal lost, flagged	26.3	25.9	−0.4	31.6	32.3	+0.7
Visible dark	30.0	28.2	−1.8	39.7	36.6	−3.1
Thermal noise	51.6	34.1	−17.5	75.4	45.9	−29.5
Tracking, mean of four sequences						
	MOTA (%)			IDF1 (%)		
Condition	Image	Memory	Change	Image	Memory	Change
Clean	44.5	42.5	−2.0	64.3	61.3	−3.0
Thermal lost	35.3	34.7	−0.6	53.6	50.7	−2.9

in Section 5.1 and because both models are nano scale and trained for 25 epochs on a subsample.

5.3. Temporal State Memory

The memory is trained for 3 epochs on clips of 6 frames with the detector frozen (batch 4, learning rate 5×10^{-4}), and is attached to the CFFM-Net v2 trained on KAIST. All rows of Table 10 use that one memory.

On clean video the memory changes the miss rate over all frames from 20.3 % to 19.8 % (−0.5 points) and AP_{50} by +0.5 points. This is a gain of half a point from one run, and we do not treat it as established.

Under failures the picture depends on the fault. When the thermal camera is lost the memory does not help where it should: over all frames the miss rate moves by −0.3 points, but on the failed frames it is 32.3 % with memory against 31.6 % without (+0.7 points, slightly worse), and at night the all-frame result is also worse (+0.9 points). Declaring the lost camera through the sensor flag leaves this unchanged (+0.7 points on failed frames). The all-frame improvement comes from the frames outside the failures, where the miss rate falls from 19.9 % to 18.6 %. A lost thermal camera is the case a recurrent state should ride out, and a four-second outage appears to be too long for this memory to bridge,

though we did not test other outage lengths. When the visible camera is darkened the memory helps on the failed frames, 39.7 % to 36.6 % (−3.1 points), with the largest part of the gain at night (−4.8 points over all night frames and −0.4 by day).

The large effect is under thermal noise. Without memory, noise at severity three on the thermal camera drives the miss rate from 20.3 % to 51.6 %, and to 75.4 % on the failed frames; with memory it is 34.1 % overall and 45.9 % on failed frames, a change of −17.5 and −29.5 points, and AP_{50} rises from 50.7 % to 72.7 %. This comparison has a confound that we cannot remove with the runs we have. The memory was trained on clips with injected faults, noise among them, while the detector it sits on was not, so the gain may come from the memory block learning to suppress noisy features rather than from carrying state across frames. A control that trains a stateless adapter of the same size on the same clips would separate the two, and we have not run it. The result shows that the memory block can recover most of the accuracy lost to thermal noise; it does not show that the recurrent state is the reason.

5.4. Tracking

The memory does not improve tracking in our runs (Table 10). On clean video MOTA falls from 44.5 % to 42.5 % and IDF1 from 64.3 % to 61.3 %, and the memory is ahead on MOTA in 1 of 4 sequences. With the thermal camera failing, the cost of the failure itself is large without memory (MOTA down by 9.1 points), and memory recovers none of it: MOTA is 34.7 % against 35.3 %, IDF1 is 50.7 % against 53.6 %, and the memory wins MOTA on 2 of 4 sequences. Identity switches are close (33 against 35 on clean video and 38 against 41 with the failure, summed over the four sequences). Four sequences give a coarse estimate, but the sign of the IDF1 change is the same in both conditions. We therefore report the memory as a detection-time component whose benefit shows under visible darkness and thermal noise, and do not claim a tracking benefit.

Supplementary material. Supplementary video S1 (`failure_memory.mp4`, ten seconds at ten frames per second) shows a night sequence at the moment the thermal camera is cut, with the image-only detector above and the memory detector below.

6. Live Runtime Experiments

This section tests the runtime of Section 3.9 on the two properties a deployed system needs: that delay and memory stay bounded over a long run, and that a failing camera is noticed quickly and without false alarms. Recorded KAIST test sequences are replayed at real speed as the two cameras, so every run is repeatable and the faults are exact. The runtime uses CFFM-Net v2 trained on KAIST with the memory of Section 5.3 and ByteTrack. It ran on the same RTX 4050 Laptop GPU as Table 6; its inference times are higher than there because they are measured inside the running pipeline and include the memory, pre-processing and the tracker.

Table 11: Live runtime on replayed KAIST sequences. Soak: paired cameras at the stated rate. Link faults: 20 fps with a scripted cable pull, freeze and stall. Drift is the late minus the early median end-to-end delay. One run per row.

	Soak		Link faults	
Measure	30 fps	20 fps	Monitor on	Monitor off
Run time (s)	594	117	107	107
Throughput (frames/s)	12.4	12.7	11.8	12.0
Capture drop (%)	58.8	36.3	–	–
End-to-end median (ms)	94.4	100.0	101.5	100.7
End-to-end p95 (ms)	117.9	129.9	139.4	139.4
Inference median (ms)	70.9	68.3	72.6	70.5
Inference p95 (ms)	88.7	89.3	98.6	99.9
Drift (ms)	+0.3	+8.2	+10.7	−2.8
GPU memory growth (MB)	−1	0	+1	−1
Miss rate, all frames (%)	–	–	27.9	27.1
Camera reconnects	–	–	10	10

6.1. Soak test

Paired 640×512 cameras ran at 30 frames per second for ten minutes (594 s) and at 20 frames per second for 120 s. A frame that the model cannot take is dropped at capture and never queued.

Table 11 gives the results. The stability criterion is met: over the ten-minute run the end-to-end delay drifts by +0.3 ms and GPU memory changes by −1 MB, so neither grows. The runtime is not real time at the camera rate on this hardware, however. At a nominal 30 frames per second it processes 12.4 and drops 58.8 % of the captured frames, and at 20 frames per second it processes 12.7 and drops 36.3 %. The median inference time with memory and tracking is 70.9 ms and the median end-to-end delay 94.4 ms at 30 frames per second, with a 95th percentile of 117.9 ms. The delay stays bounded because the capture drops frames rather than queueing them, which trades temporal resolution for latency. The shorter run drifts by +8.2 ms, about eight percent of its median delay, and is too short to show whether that settles.

6.2. Link faults and the stream monitor

A scripted run of 110 s at 20 frames per second applies three faults: the thermal cable pulled for 6 s from second 20, the visible camera frozen for 6 s from second 50, and the thermal stream stalled for 3 s from second 80. Throughout, 10 % of frames are dropped at random and the thermal stream is delayed by 60 ms. We repeat the run with the monitor feeding the sensor flags to the model (on) and with the monitor observing only (off).

Table 12 gives the monitor timing. A cable pull is flagged 0.29 s after onset and cleared 0.19 s after the camera returns, a stall is flagged after 0.31 s and cleared after 0.12 s, and a frozen visible camera is flagged after 0.69 s, which is longer because a freeze is recognised only after consecutive nearly identical frames and a replayed stream drops frames at random. The monitor is not free of false alarms: the thermal stream is marked down outside the scripted

Table 12: Stream monitor timing for each scripted fault, with the monitor feeding the model (on) and only observing (off). Delays are seconds from onset or end of the fault to the first change of state.

	Detect (s)		Recover (s)		False alarms (%)	
Fault	On	Off	On	Off	On	Off
Thermal unplugged, 6 s	0.29	0.28	0.19	0.18	4.3	5.0
Visible frozen, 6 s	0.69	0.52	0.10	0.13	0.0	0.0
Thermal stalled, 3 s	0.31	0.26	0.12	0.16	6.1	8.0

windows in 4.3 % of records in the cable-pull measurement and 6.1 % in the stall measurement, which we attribute to the random drops and the added delay but did not test separately. The visible camera had none.

Feeding the flags to the model gives no measured accuracy benefit. The miss rate over all frames is 27.9 % with the monitor on and 27.1 % with it off, a difference of +0.8 points against the monitor, and throughput and delay are the same within a fraction of a frame (11.8 against 12.0 frames per second). This is one run, the scripted faults cover only fifteen of the 110 seconds, and the memory-trained detector may already cope with a black thermal frame, in line with the flagged-versus-unflagged result of Section 4.4. We conclude that the monitor detects and clears link faults quickly, and that we have not shown that its flags improve detection. Both runs reopened the cameras 10 times.

Supplementary material. Supplementary video S2 (`live_session.mp4`, about nine seconds at 20 frames per second) shows the live board during a thermal cable pull: both camera images, tracked and numbered subjects, the health of each link and the delay of each frame.

7. The Full System

The detector is one stage of a pipeline that runs from cameras to counts. Two paired cameras feed the live runtime of Section 3.9, whose monitor converts stream health into sensor flags. CFFM-Net with its Temporal State Memory produces detections, ByteTrack turns them into tracks, and records of latency, health and track state feed the analysis boards. A separate layer takes detections from several calibrated cameras and produces anonymous ground-plane tracks and aggregate analytics. This section describes that layer and reports its results.

7.1. Multi-camera layer

The multi-camera layer uses a plain RGB YOLO26-m detector, not CFFM-Net. WILDTRACK [43] has seven synchronised, calibrated visible cameras and no thermal channel, so the layer demonstrates the system's last stage, from boxes to analytics, and provides no evidence about fusion. We evaluate the same layer with the detector zero-shot from COCO weights and fine-tuned on the training frames, which are exported as per-view person boxes. Frames are sampled every 5 frames, and the frames from index 1800 on are held out of the export.

From boxes to the ground plane. Camera calibration gives a homography between the ground plane, in centimetres, and each image. The bottom centre of each person box is mapped to the ground, and points that fall outside the monitored area or whose back-projected ray misses the ground in front of the camera are discarded. Points from different cameras are merged by visibility-normalised voting. A candidate position is kept only if enough of the cameras that could see a person standing there did detect one near it, which avoids penalising people who stand in a blind spot of most cameras. Each candidate must also have enough support: the mean of the best nearby scores, normalised by the number of cameras that see the position, must exceed a threshold, and candidates closer than a minimum separation are suppressed in favour of the more confident. The merging radius is 40 cm for both detectors, the required fraction of visible cameras is 0.30, and the score threshold is 0.45 for the zero-shot detector and 0.30 for the fine-tuned one.

Tracking and analytics. Fused positions feed a constant-velocity tracker on the ground plane with Hungarian association, a gate of 75 cm, a track kept through 3 missed frames, and confirmation after 2 hits. Identifiers are ephemeral and anonymous. The analytics are aggregate: counts per zone and in total, crossings of a line in each direction, dwell time per zone, and an occupancy map on a grid of 25 cm cells. As a privacy measure, the head region of every person box is blurred before any image is stored or displayed.

7.2. Results

Table 13 reports detection, tracking and counting. Zero-shot, the detector has high precision (92.1 %) and low recall (46.5 %), giving a MODA of 42.5 %. These are the numbers of a detector that has never seen the viewpoint: it finds 443 people and misses 509. Fine-tuning on the training frames lifts MODA to 77.4 %, with recall 87.3 % and precision 89.8 %, and cuts the misses to 121 at the price of more false positives (94 against 38). Localisation quality, MODP, is similar for the two, 77.6 % and 76.5 %.

On the ground-plane tracks the system reaches MOTA 79.4 % and IDF1 84.9 %, with 18 identity switches, and tracks 25 of 41 people for most of their presence. Counting is accurate enough to be useful: the mean absolute error of the total count is 1.43 people on a mean of 23.8 people per frame, a relative error of 6.0 %, and the error per zone ranges from 0.43 to 1.18 people. Crossings of the reference line number 11 in one direction and 0 in the other. The busiest zone is the second, with 46 tracks and a mean dwell of 12.4 s, against 5.7 s in the first and 4.0 s in the third.

These results describe the multi-camera layer on RGB input. Tracking and counting run on the fine-tuned YOLO26-m detections. They say nothing about whether visible-thermal fusion improves analytics, which would require paired multi-camera thermal data that WILDTRACK lacks.

8. Discussion

What the evidence supports. Fusing the two cameras helps on M3FD: against the matched visible-only detector, which is the stronger single camera there, CFFM-Net v2 gains +4.0 ± 0.1 mAP and the concat baseline +3.4 ± 0.3 mAP,

Table 13: Multi-camera analytics. Detection uses zero-shot and fine-tuned YOLO26-m with ground-plane fusion; tracking and counting are reported once.

Measure	Zero-shot	Fine-tuned
Detection		
MODA (%)	42.5	77.4
MODP (%)	77.6	76.5
Precision (%)	92.1	89.8
Recall (%)	46.5	87.3
True positives	443	831
False positives	38	94
False negatives	509	121
Tracking		
MOTA (%)	79.4	
IDF1 (%)	84.9	
ID switches	18	
Mostly tracked of 41	25	
Counting		
Count MAE (people)	1.43	
Relative error (%)	6.0	
Zone one MAE	0.93	
Zone two MAE	1.18	
Zone three MAE	0.43	

with the caveat that the single-camera rows are one seed. The fuser design adds a smaller step: v2 beats the matched concat baseline on M3FD by $+0.6 \pm 0.4$ mAP with the same sign on all three seeds, but the margin is not significant at $n = 3$ (paired $p = 0.11$; one-sided sign test 0.125), and the small-object margin of $+0.4 \pm 0.3$ pp is not significant either. v2 costs 1.36 times the latency, 1.48 times the parameters and 1.70 times the FLOPs of the concat baseline, which passes the limit of at most twice the baseline latency that was fixed in advance. Under sensor faults on M3FD, v2 is ahead of the concat baseline on thermal-side faults by $+3.6 \pm 0.2$ pp and behind on visible-side faults by -0.9 ± 0.9 pp, a pattern consistent across three seeds but not tested for significance per fault. The video and runtime experiments support three narrower statements. The Temporal State Memory lowers the KAIST miss rate under thermal noise (51.6 % to 34.1 %) and darkness of the visible camera (30.0 % to 28.2 %). The live runtime keeps delay and GPU memory bounded over ten minutes. And its stream monitor flags a cable pull within 0.29 s.

What it does not support. The evidence does not show that CFFM-Net is better on LLVIP, where the margin of +0.8 pp has a spread of 1.1 pp, or on KAIST stills, where the baseline has the lower miss rate (19.8 % against 20.3 %). It does not show that the gate makes the model more robust than simpler means: a concat baseline trained with fault augmentation retains 86.8 % of its clean mAP on M3FD against 76.3 % for v2, and v2 loses heavily to noise faults on

LLVIP. It does not show that the reliability branch helps: in a preliminary single-seed run v3 is worse than v2 and than v3 without the branch on both datasets. It does not show that the selective scan, as opposed to the other changes made between round one and v2, is the source of the gain, since the round-one controls with a gated convolution of matched cost land within one seed's noise of the scan. It does not show that the memory gain under thermal noise comes from the recurrent state, because the memory was trained on faulty clips and the detector under it was not, and we ran no stateless adapter as a control. It does not show a tracking benefit from the memory, a benefit from feeding the monitor flags to the model (+0.8 points of miss rate in a single run), or real-time operation at the camera rate, since the runtime processes 12.4 of a nominal 30 frames per second. And it does not place our models among the published state of the art: on KAIST our miss rates are above 10.62 %, the worst published figure we list, and the LLVIP and M3FD figures of Section 4.7 are in different settings.

Limits. All detectors are nano scale and trained once per seed with three seeds at most; v3 and its controls have one seed, as do the single-camera baselines and every KAIST result. The scan-versus-gated-convolution control was run at round one and has not been repeated at v2 over seeds. The fault benchmark uses synthetic faults and the test split; on LLVIP it covers one seed only, and the fault-trained baseline is scored at one seed on both datasets. The KAIST evaluation uses the original annotations and our own implementation of the miss rate, so it is not comparable with published figures. The live runtime was run on replayed recordings on the same laptop GPU, not on physical cameras, and its fault script covers three events in one run. Latency in Table 6 was measured on one RTX 4050 Laptop GPU, so absolute values should not be compared with other papers. The multi-camera layer ran on RGB WILDTRACK, so it supports the claim that the system's last stage works, and makes no claim about fusion.

Next steps. Four experiments follow from the results. First, run the fault benchmark on a scan-based model trained with fault augmentation against the fault-trained concat baseline, at three seeds on both datasets, which decides whether the gate adds robustness once training is matched. Second, train a stateless adapter of the memory's size on the same clips, to separate the recurrent state from the extra capacity. Third, repeat the scan-versus-convolution comparison in the v2 recipe at three seeds. Fourth, find the cause of the loss to additive noise on LLVIP, whose gate has no cue that separates noise from texture.

9. Conclusion

CFFM-Net places the decision about camera trust where a state-space model writes its state: a per-pixel reliability for each camera scales the step of a four-direction scan over interleaved visible and thermal tokens, on top of a stock YOLO26 detector. At nano scale and over three seeds, fusing the two cameras beats the best single camera on M3FD by +4.0 mAP, but most of that is available to a plain concatenation baseline, which gains +3.4 mAP. The gated scan adds $+0.6 \pm 0.4$ mAP on M3FD, not significant at $n = 3$ (paired $p = 0.11$), and nothing measurable on LLVIP or on KAIST stills. Its advantage under thermal-side faults on M3FD is consistent in direction, but a baseline trained with

fault augmentation is more robust than the model, and the reliability loss of v3 lowers accuracy. The Temporal State Memory cuts the KAIST miss rate under thermal noise from 51.6 % to 34.1 %, with a confound we could not remove, helps little when the thermal camera is lost, and does not improve tracking. The live runtime holds delay and GPU memory constant over ten minutes at 12.4 frames per second on a nominal 30, and its monitor flags a cable pull in 0.29 s. The multi-camera layer, demonstrated with an RGB-only detector on WILDTRACK, reaches 77.4 % MODA, 79.4 % MOTA and a count error of 6.0 %. The next experiment that matters is the fault benchmark on a scan-based model trained with fault augmentation against the fault-trained baseline, at three seeds on both datasets.

Declaration

No hyperparameter was tuned on a test split. The one design decision that used evaluation results was made on LLVIP at seed zero: the P2 ablations of round one led us to drop the stride-4 level in v2 (Section 4.5); these ablations were not run on M3FD, and the decision was applied unchanged to M3FD and to seeds one and two. The M3FD hold-out split is fixed and shared by every run, and the fault benchmark scores checkpoints without being used to select a model.

Data Availability

The code, configurations, notebooks and trained results are at `https://github.com/riteshroshann/yolo-cross-feature-fusion-mamba-net`, and the two supplementary videos `failure_memory.mp4` and `live_session.mp4` accompany the submission. LLVIP, M3FD, KAIST and WILDTRACK are public datasets distributed by their authors. Every number of this paper's own results is generated from the result files of that repository by a script, and the tables are built from those numbers. The published figures of Tables 7 and 8 are copied from the cited papers.

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Table A.14: Per-seed accuracy in percent, 80 epochs. The last three columns are COCO size bands.

Dataset	Model	Seed	mAP	mAP_{50}	Precision	Recall	Small	Medium	Large
LLVIP	Concat, modality dropout	0	64.8	97.2	96.8	91.7	8.0	64.7	66.1
		1	66.3	96.6	97.0	91.7	9.6	65.8	69.3
		2	65.2	96.6	96.7	90.6	11.0	64.6	68.3
	mean ± std		65.5 ± 0.8	96.8 ± 0.4	96.9 ± 0.2	91.3 ± 0.6	9.6 ± 1.5	65.0 ± 0.6	67.9 ± 1.6
	CFFM-Net v2	0	65.6	97.0	96.4	92.4	8.2	65.5	66.8
		1	66.0	96.6	97.1	91.2	9.0	65.5	68.7
		2	67.1	97.2	96.4	92.4	10.2	66.5	70.5
	mean ± std		66.2 ± 0.8	96.9 ± 0.3	96.6 ± 0.5	92.0 ± 0.7	9.1 ± 1.0	65.8 ± 0.6	68.7 ± 1.9
M3FD	Concat, modality dropout	0	53.5	82.2	85.5	76.4	34.9	67.5	87.0
		1	53.5	81.7	86.4	74.2	34.6	66.9	88.4
		2	52.9	81.8	87.6	74.3	34.8	65.9	86.2
	mean ± std		53.3 ± 0.3	81.9 ± 0.3	86.5 ± 1.1	74.9 ± 1.2	34.8 ± 0.2	66.8 ± 0.8	87.2 ± 1.1
	CFFM-Net v2	0	53.7	82.7	88.8	74.4	34.9	66.8	87.4
		1	54.0	83.4	88.7	75.0	35.1	66.5	87.8
		2	53.9	82.2	85.7	75.0	35.5	66.9	88.0
	mean ± std		53.9 ± 0.1	82.7 ± 0.6	87.7 ± 1.8	74.8 ± 0.4	35.2 ± 0.3	66.7 ± 0.2	87.7 ± 0.3

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Appendix A. Per-seed results

Table A.14 lists every run behind the headline comparison, with the mean and sample standard deviation over seeds. Table A.15 gives the per-seed mAP margin of v2 over the concat baseline, which is the quantity averaged in Table 1.

Appendix B. Fine size bands on M3FD

Table B.16 gives the finer size bands behind Section 4.3.

Table A.15: Per-seed mAP margin of CFFM-Net v2 over the concat baseline, in percentage points.

Dataset	Seed 0	Seed 1	Seed 2
LLVIP	+0.7	−0.4	+1.9
M3FD	+0.3	+0.5	+1.0

Table B.16: M3FD average precision in the finer size bands, percent, per seed and as mean and sample standard deviation.

Model	Seed	Very tiny	Tiny	Small
Concat, modality dropout	0	7.5	24.7	48.6
	1	7.8	25.5	47.3
	2	7.6	21.6	49.2
	mean ± *std*	7.6 ± 0.2	23.9 ± 2.1	48.4 ± 1.0
CFFM-Net v2	0	7.6	24.7	47.9
	1	7.7	24.9	48.4
	2	8.9	25.3	48.6
	mean ± *std*	8.0 ± 0.8	24.9 ± 0.3	48.3 ± 0.4

Appendix C. Degradation probe of the round-one models

Table C.17 gives the LLVIP degradation probe that preceded the fault benchmark. It uses the round-one models at 30 epochs and one seed.

Table C.17: Degradation probe on LLVIP, mAP in percent, round-one models (30 epochs, seed 0).

Condition	v1	v1, no gate	Concat
Clean	62.9	62.6	65.0
Visible dark	59.9	54.2	61.0
Visible drop	56.7	57.0	61.8
Visible drop, flagged	59.5	57.0	61.8
Thermal drop	2.2	0.7	0.3
Thermal drop, flagged	17.4	0.7	0.3
Thermal shift 4 px	53.3	51.8	51.6
Thermal shift 8 px	32.6	31.1	29.7
Thermal shift 16 px	2.8	2.9	1.7

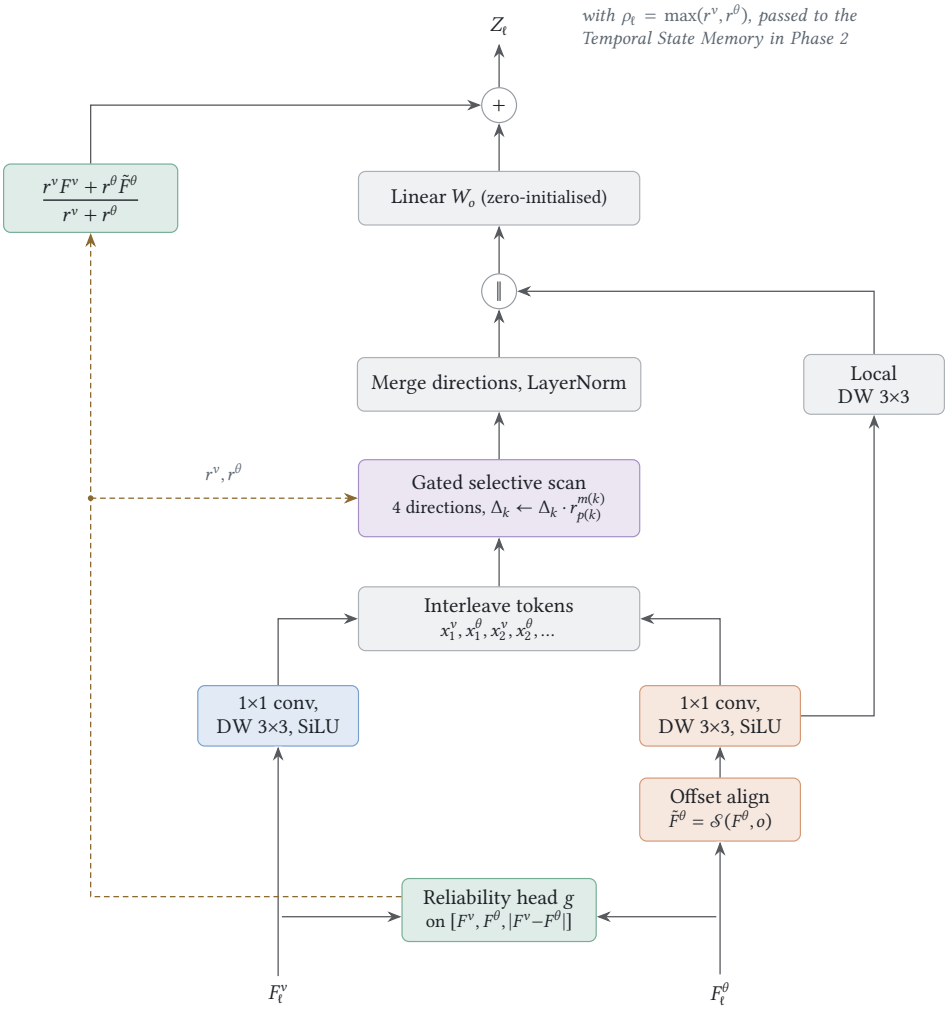


Figure 3: The CMFM block. A reliability head weights each camera, thermal features are warped onto the visible grid, visible and thermal tokens are interleaved and scanned in four directions with the step scaled by reliability, and a local convolution branch joins before a zero-initialised projection. The result adds to a reliability-weighted average. The larger of the two reliabilities is also passed to the temporal memory.

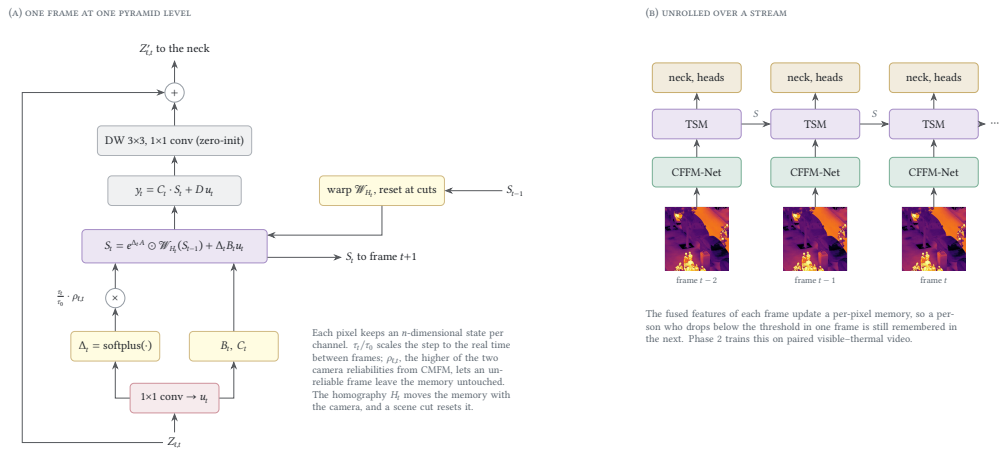


Figure 4: The Temporal State Memory. (a) One level at one frame: the step is scaled by elapsed time and by the camera reliability, and the previous state is warped by the frame-to-frame homography. (b) Unrolled over a stream, the memory sits between the fusion output and the neck.

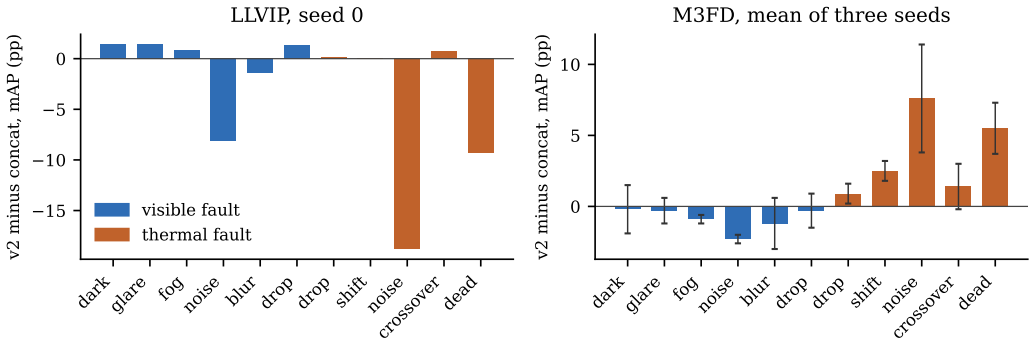


Figure 5: Margin of CFFM-Net v2 over the concat baseline for each fault at severity three, in mAP points, from Table 3. Positive means v2 is ahead. LLVIP is one seed; M3FD shows the mean over three seeds with the sample standard deviation as an error bar.



Figure 6: A LLVIP test pair as the round-one CFFM-Net sees it: visible with detections, thermal, and the fusion trust map in which blue marks visible and orange marks thermal weight.

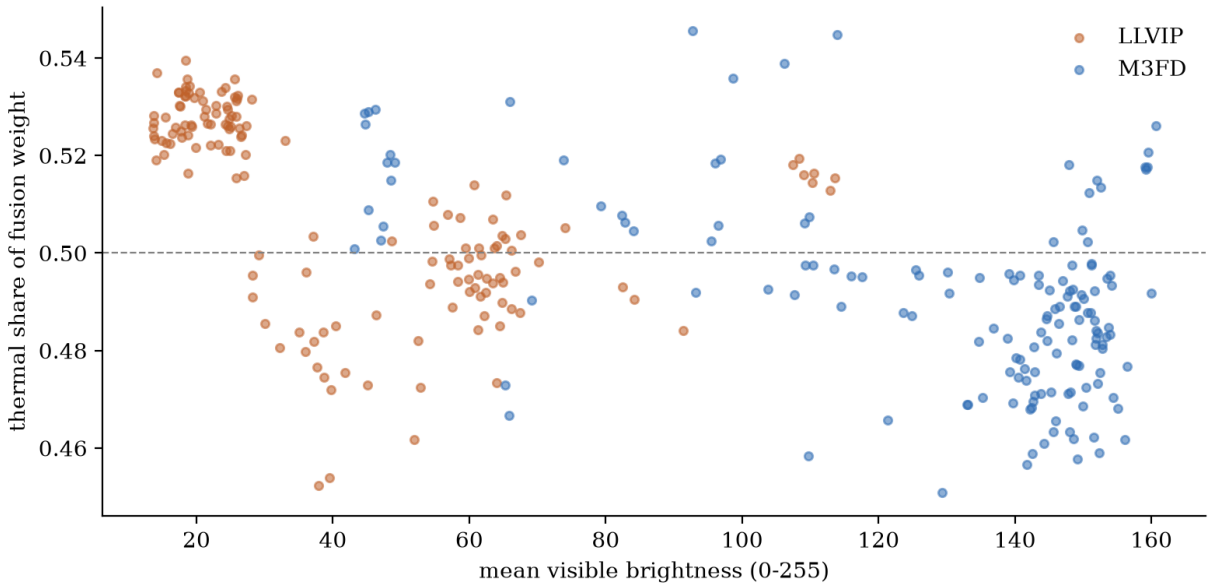


Figure 7: Thermal share of the fusion weight against mean visible brightness for LLVIP and M3FD test images, with a round-one CFFM-Net. The dashed line marks an equal share.